



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term Examination, _____ Semester

Course: CSE207 Data Structures, Section-
Instructor: Puja Chakraborty, Lecturer, CSE Department, EWU.
Full Marks: 30
Time: 1 hour 10 mins

Note: There are SIX questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

- 1) Fig-01 contains the representation of a linked list in the memory. Draw a schematic diagram of the linked list from Fig-01.

[CO1, C3,
Mark: 5]

	INFO	NEXT
1	L	13
2	-	4
3	0	10
4	C	14
5	e	2
6	-	1
7	v	5
8	2	3
9	E	8
10	7	x
11		
12	I	6
13	o	7
14	S	9

Start
↓
→

Fig-01: Linked List Representation

- 2) Show the steps to calculate the value from the reverse polish notation given below using a stack:

$$a \ b \ c \ + \ d \ * \ + \ e \ -$$

Here,

a: last digit of your student id

b: 2

c: 3

d: last digit of your student id

e: 2

[CO3, C3,
Mark: 5]

- 3) Write the pseudocode to insert a new element to a stack.

Algo 3.1: push(*top*: pointer to the last element of the stack, *len*: no of currently available elements, *maxLen*: no of maximum elements allowed in the stack, *key*: element to insert)

[CO3, C3,
Mark: 5]